

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path,'NG_FFCutsFits.mat');

if ~exist('NG_FFCutsFits','var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(NG_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off200NGminFF'

```

```

selectCuts =
'Ei800_off200NGminFF'

```

```

if contains(selectCuts,'msk')
    error('Can not fit field: %s',selectCuts);
end
if contains(selectCuts,'all_fit_par')
    error('Can not fit field: %s',selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei800_off200NGminFF

```

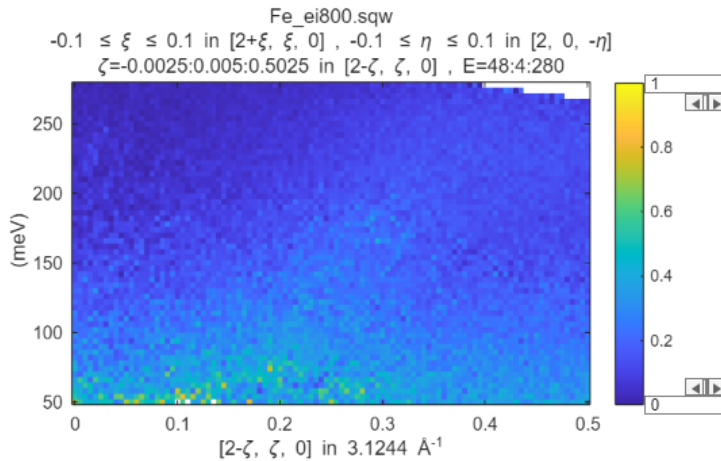
```

%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

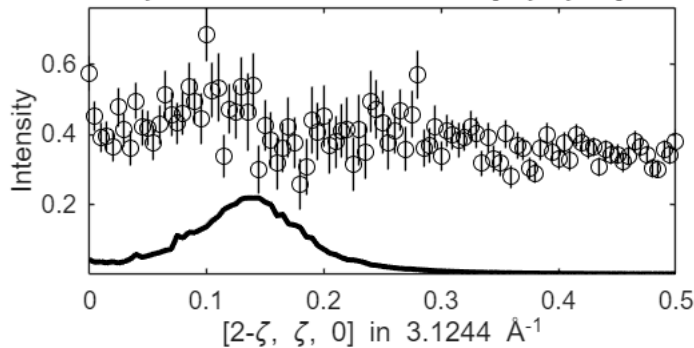
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

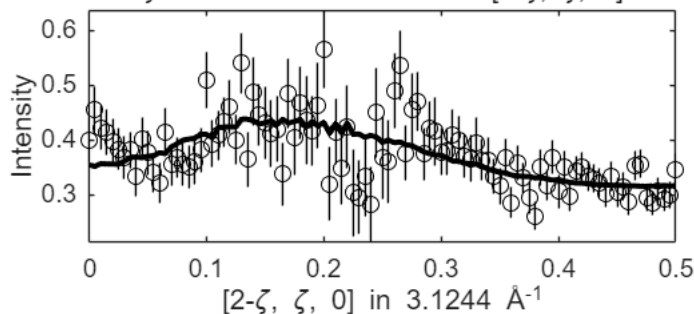
```
*****
**** En = 53±5 going to 273
*****
```

Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 48
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

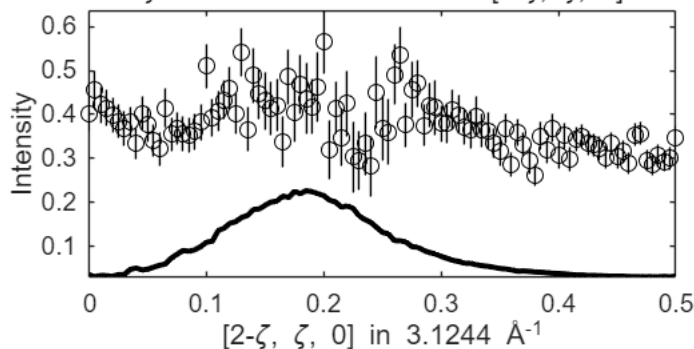


 **** En = 63±5 going to 273

Fe_ei800.sqw
 ± 11 ; J0= $46 \pm 1.4 \times 10^2$; J1= 0 ± 0 ; gamma=165.706
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 58
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



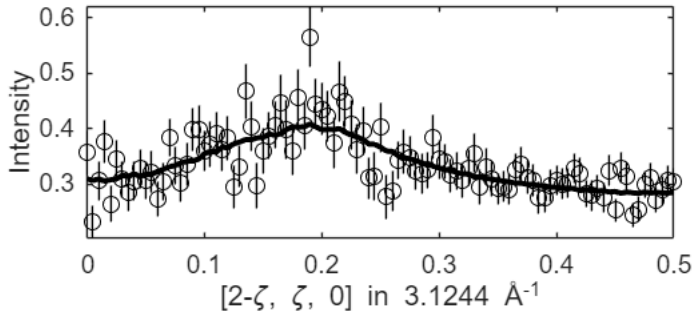
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 58
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



 **** En = 73±5 going to 273

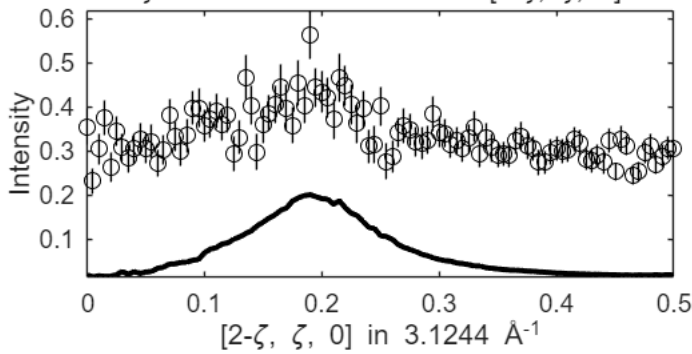
Fe_ei800.sqw

2 ± 0.084 ; $J_0 = 32 \pm 1.2$; $J_1 = 0 \pm 0$; $\gamma = 60.3914 \pm 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 68
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

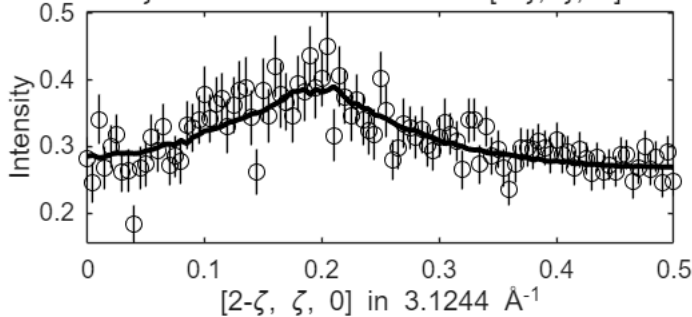
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 68
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



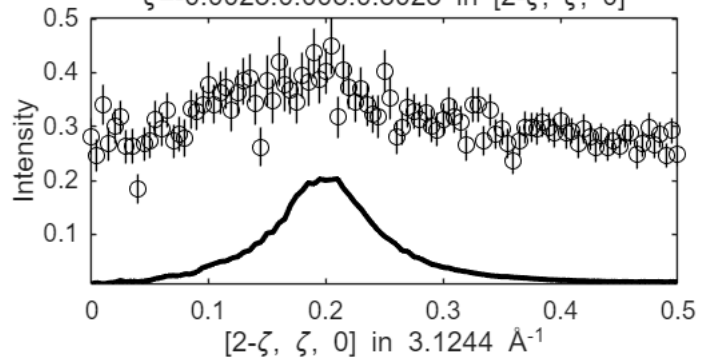
 **** En = 83 ± 5 going to 273

Fe_ei800.sqw

6 ± 0.048 ; $J_0 = 29 \pm 1$; $J_1 = 0 \pm 0$; $\gamma = 34.9327 \pm 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 78
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

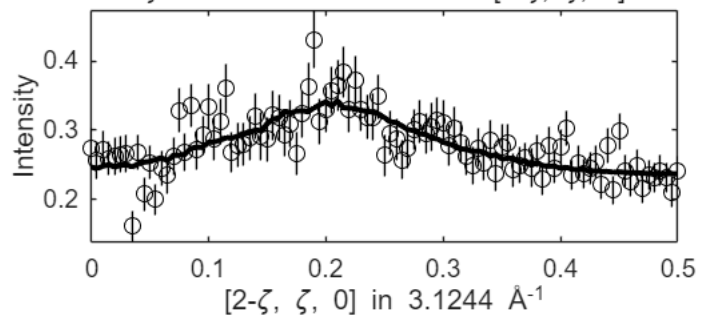


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 78
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

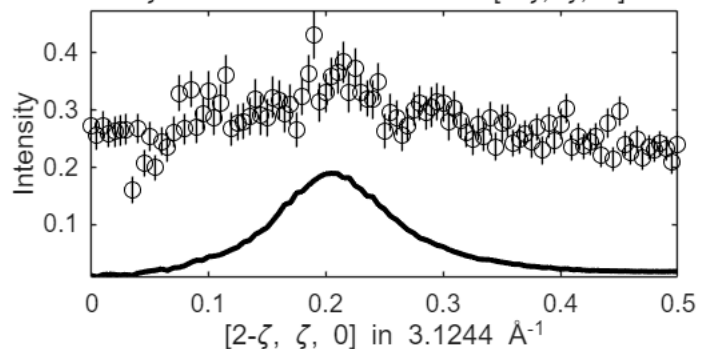


**** En = 93±5 going to 273

Fe_ei800.sqw
 1 ± 0.58 ; $J_0 = 34 \pm 6.6$; $J_1 = 0 \pm 0$; $\gamma = 55.2635 \pm 4$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 88
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



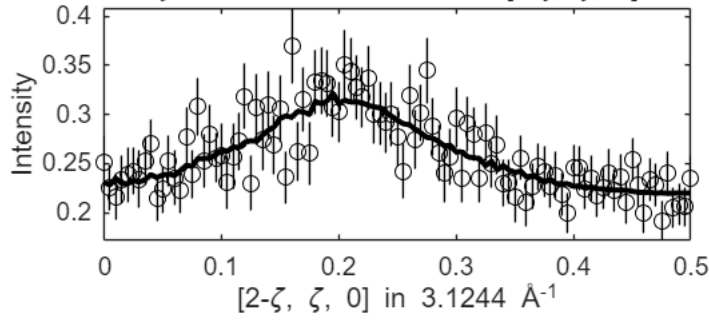
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 88
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



**** En = 103±5 going to 273

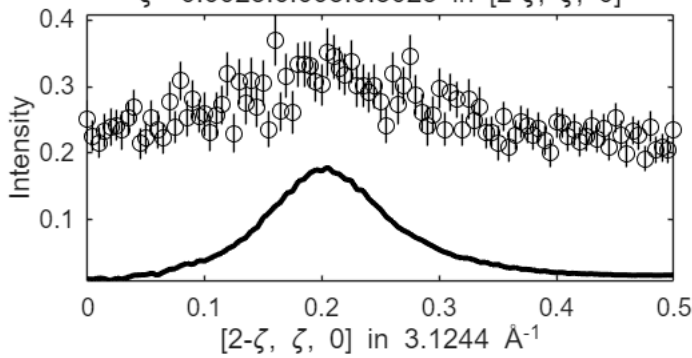
Fe_ei800.sqw

1.1 ± 0.57 ; $J_0 = 38 \pm 7$; $J_1 = 0 \pm 0$; $\gamma = 64.275 \pm 4$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 98
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

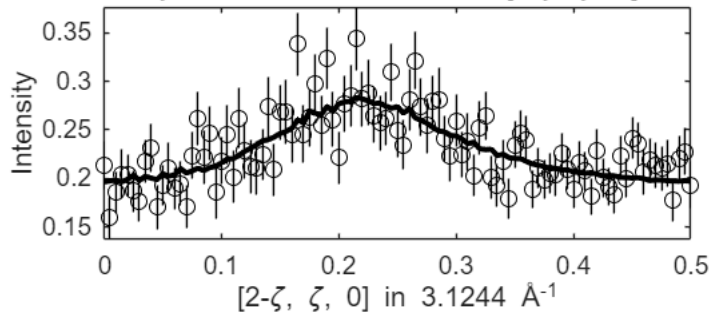
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 98
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



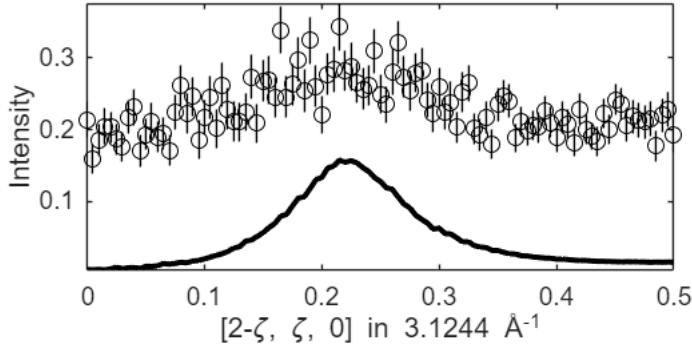
 **** En = 113 ± 5 going to 273

Fe_ei800.sqw

7 ± 0.051 ; $J_0 = 34 \pm 1$; $J_1 = 0 \pm 0$; $\gamma = 50.6635 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 108
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

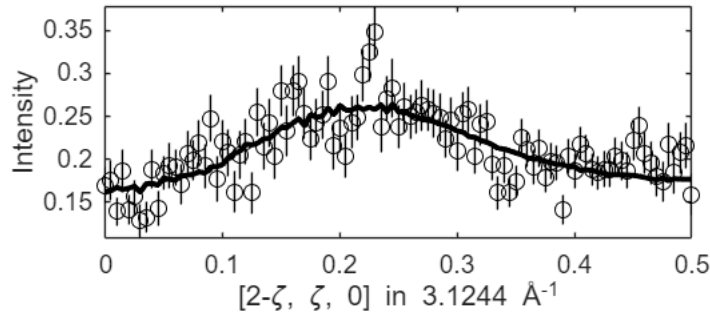


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 108
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

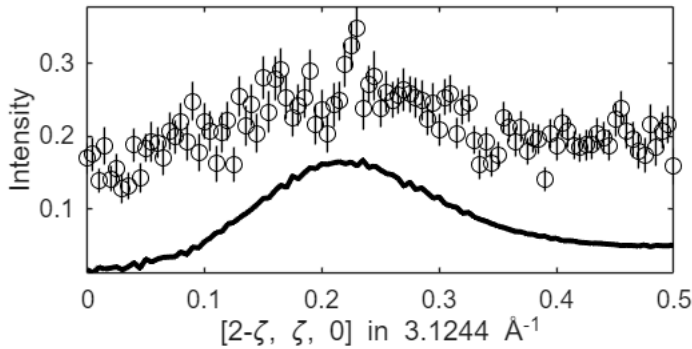


 **** En = 123±5 going to 273

Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 118
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



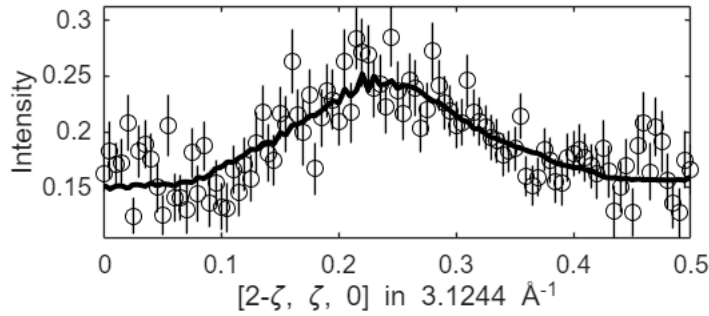
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 118
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



 **** En = 133±5 going to 273

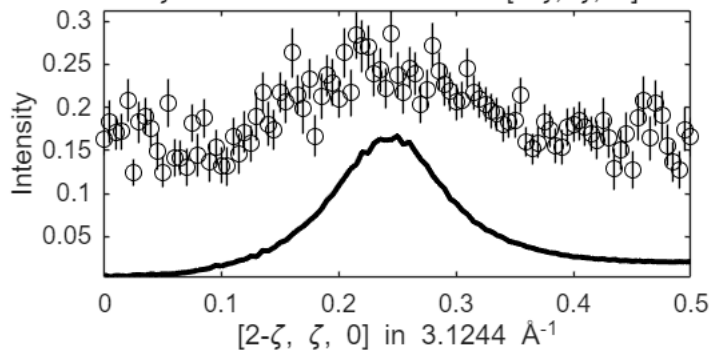
Fe_ei800.sqw

1 ± 0.046 ; $J_0 = 35 \pm 0.79$; $J_1 = 0 \pm 0$; $\gamma = 51.9219 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 128
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

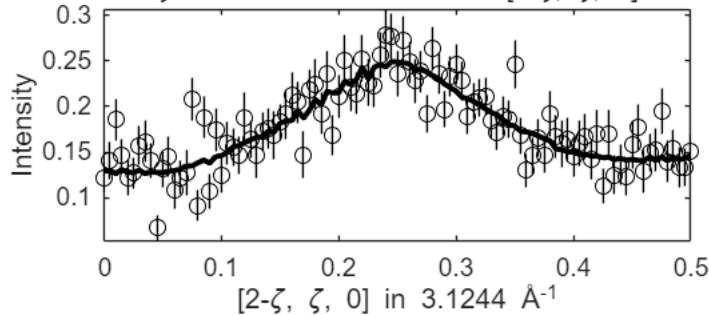
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 128
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



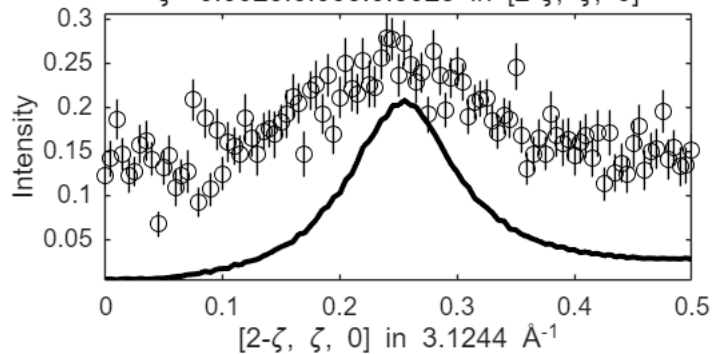
 **** En = 143 ± 5 going to 273

Fe_ei800.sqw

94 ± 0.22 ; $J_0 = 34 \pm 2$; $J_1 = 0 \pm 0$; $\gamma = 49.071 \pm 1$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 138
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

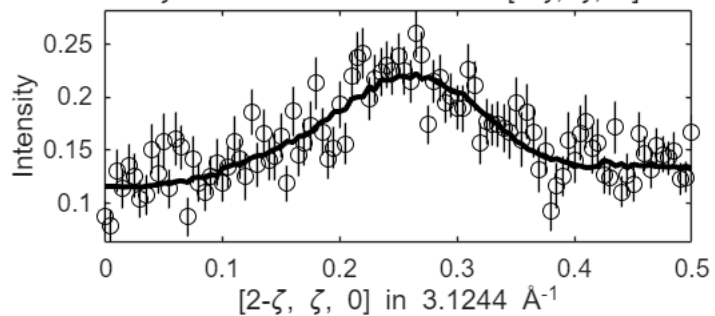


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 138
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

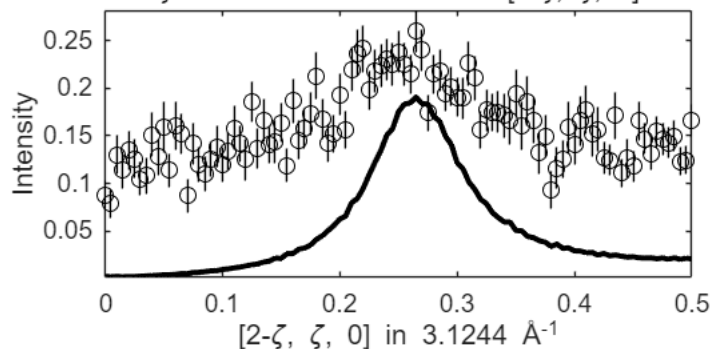


**** En = 153±5 going to 273

Fe_ei800.sqw
 1 ± 0.036 ; $J_0 = 34 \pm 0.59$; $J_1 = 0 \pm 0$; $\gamma = 39.3194 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 148
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



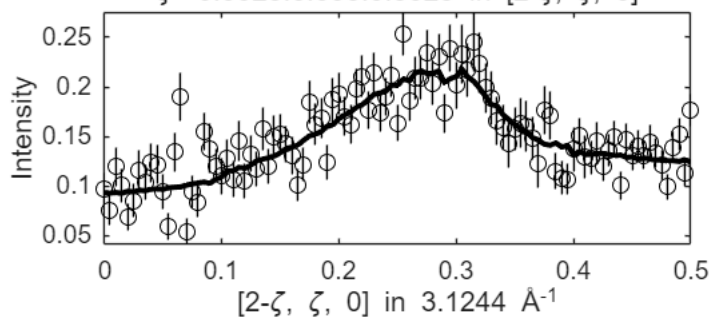
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 148
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



**** En = 163±5 going to 273

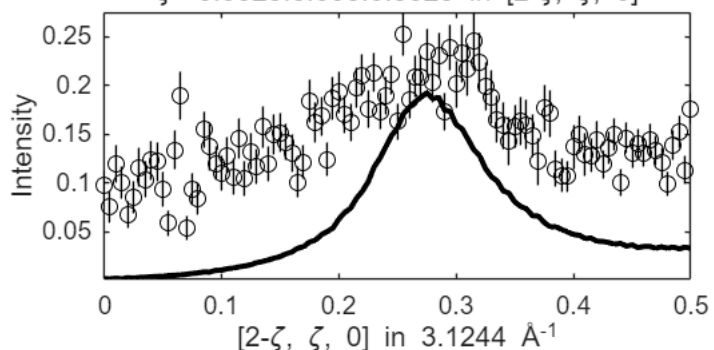
Fe_ei800.sqw

86 ± 0.05 ; $J_0 = 35 \pm 0.51$; $J_1 = 0 \pm 0$; $\gamma = 49.1239 \pm 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 158
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

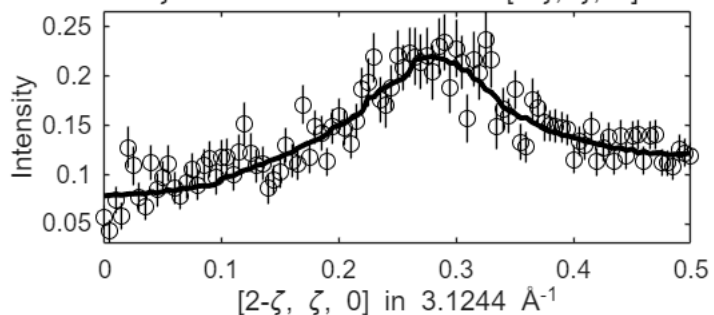
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 158
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



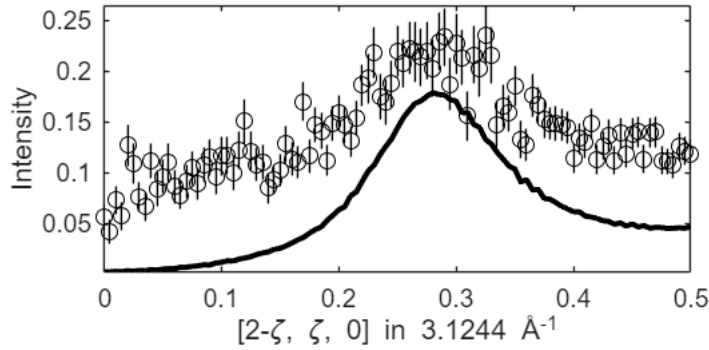
 **** En = 173±5 going to 273

Fe_ei800.sqw

1 ± 0.05 ; $J_0 = 37 \pm 0.38$; $J_1 = 0 \pm 0$; $\gamma = 63.1704 \pm 1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 168
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

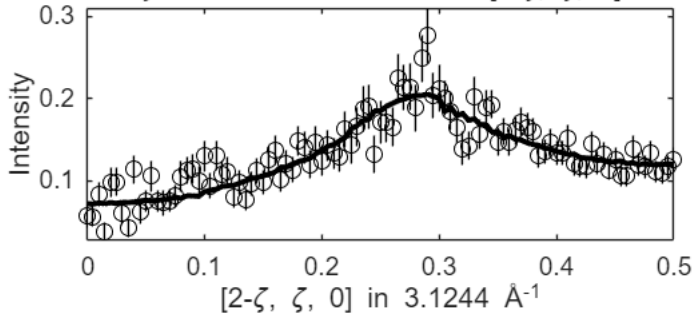


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 168
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

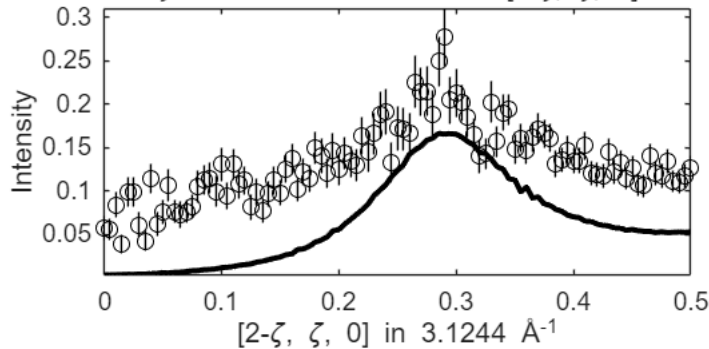


 **** En = 183±5 going to 273

Fe_ei800.sqw
 1 ± 0.054 ; $J_0 = 37 \pm 0.39$; $J_1 = 0 \pm 0$; $\gamma = 67.8358 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 178
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



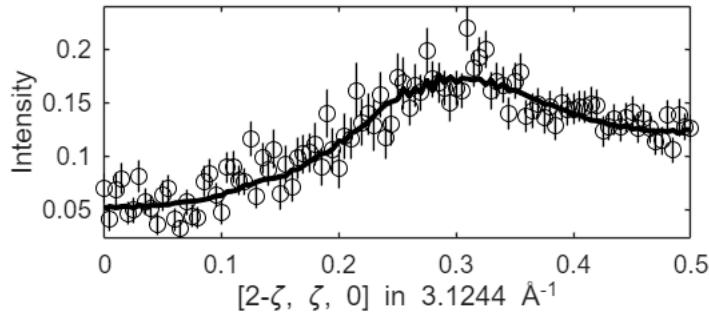
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 178
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



 **** En = 193±5 going to 273

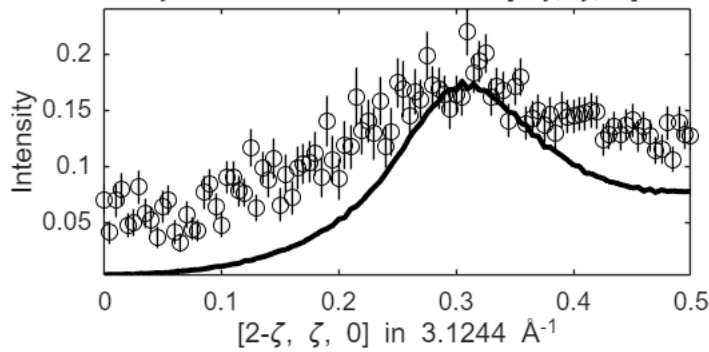
Fe_ei800.sqw

1.1 ± 0.056 ; $J_0 = 37 \pm 0.37$; $J_1 = 0 \pm 0$; $\gamma = 77.32 \pm 2$.
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 188
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

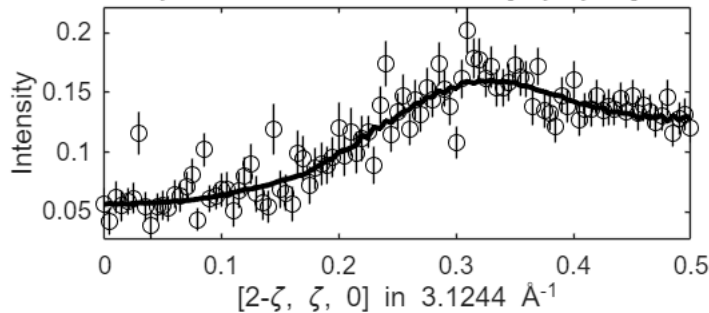
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 188
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



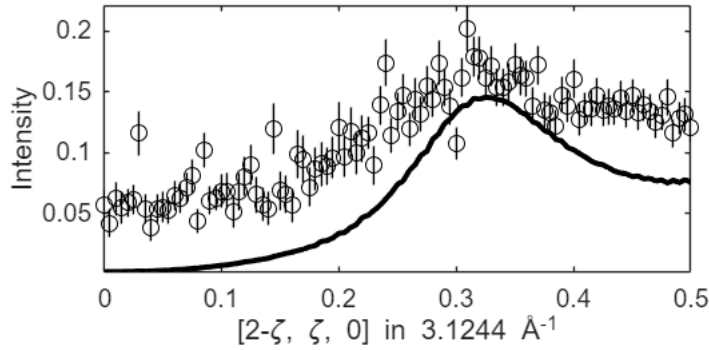
 **** $E_n = 203 \pm 5$ going to 273

Fe_ei800.sqw

38 ± 0.039 ; $J_0 = 35 \pm 0.36$; $J_1 = 0 \pm 0$; $\gamma = 71.568 \pm 1$.
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 198
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

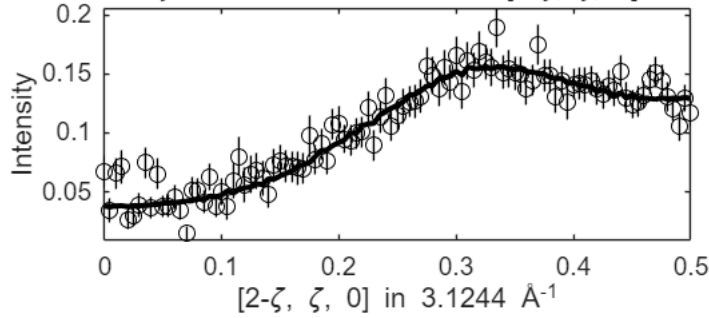


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 198
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

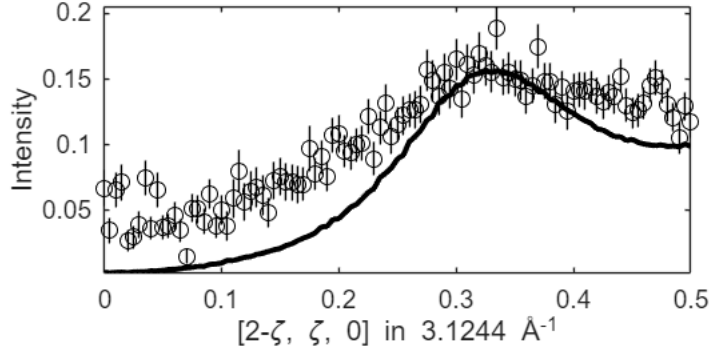


 **** En = 213±5 going to 273

Fe_ei800.sqw
 2 ± 0.061 ; $J_0 = 38 \pm 0.45$; $J_1 = 0 \pm 0$; $\gamma = 96.5695 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 208
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



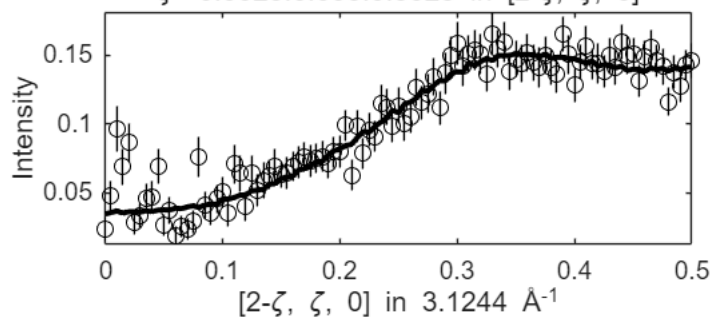
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 208
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



 **** En = 223±5 going to 273

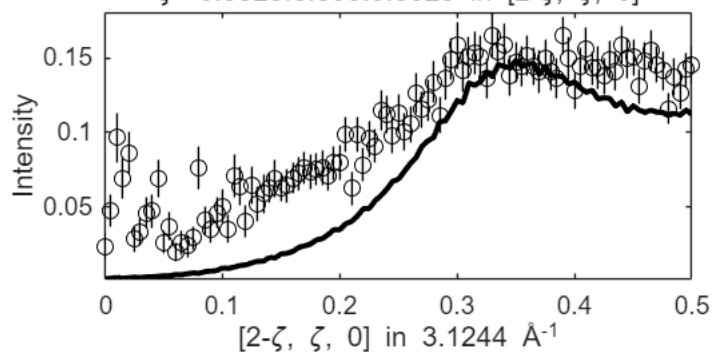
Fe_ei800.sqw

1 ± 0.057 ; $J_0 = 37 \pm 0.5$; $J_1 = 0 \pm 0$; $\gamma = 99.0618 \pm 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 218
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

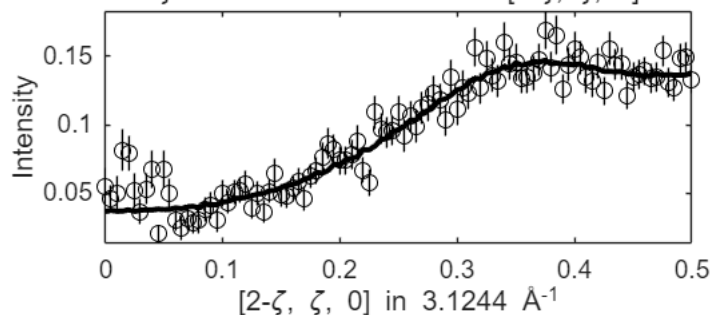
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 218
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



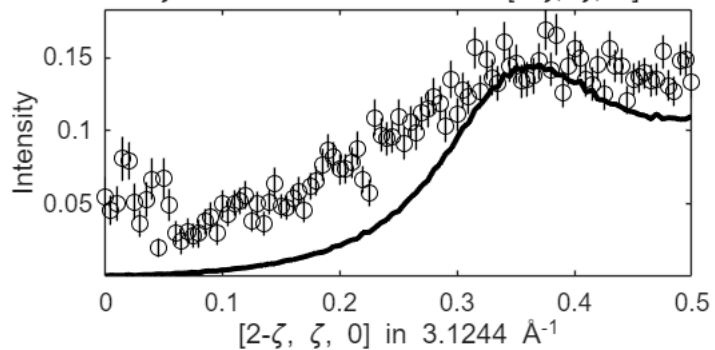
 **** En = 233±5 going to 273

Fe_ei800.sqw

5 ± 0.037 ; $J_0 = 36 \pm 0.38$; $J_1 = 0 \pm 0$; $\gamma = 73.3039 \pm 0.1$ in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 228
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

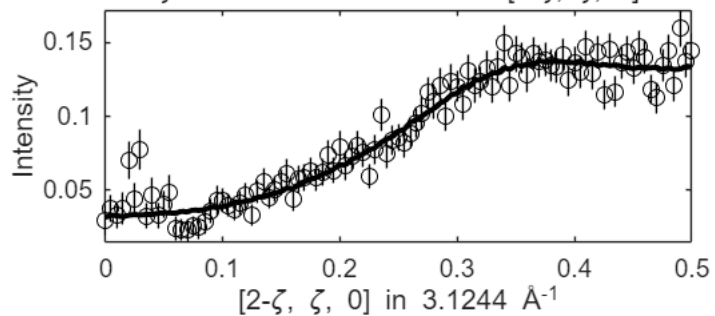


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 228
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

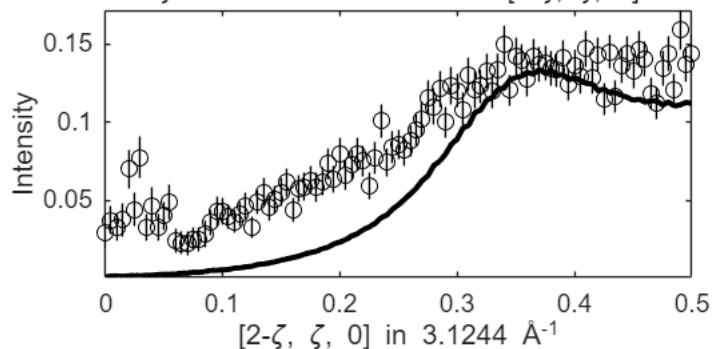


**** En = 243±5 going to 273

Fe_ei800.sqw
 2 ± 0.038 ; $J_0 = 37 \pm 0.42$; $J_1 = 0 \pm 0$; $\gamma = 88.4629 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 238
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



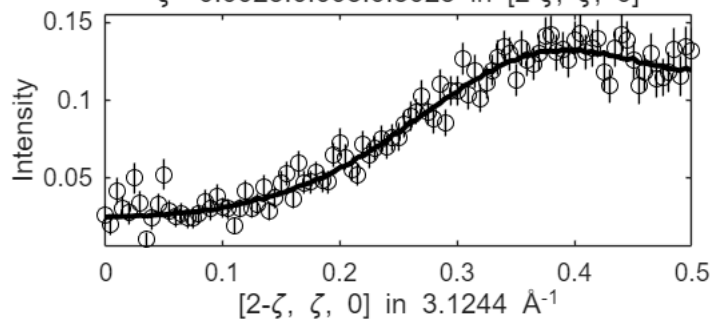
Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 238
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



**** En = 253±5 going to 273

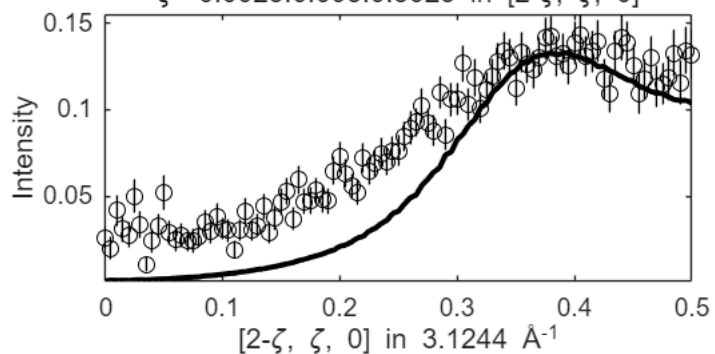
Fe_ei800.sqw

1 ± 0.035 ; $J_0 = 38 \pm 0.39$; $J_1 = 0 \pm 0$; $\gamma = 85.2643 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 248
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw

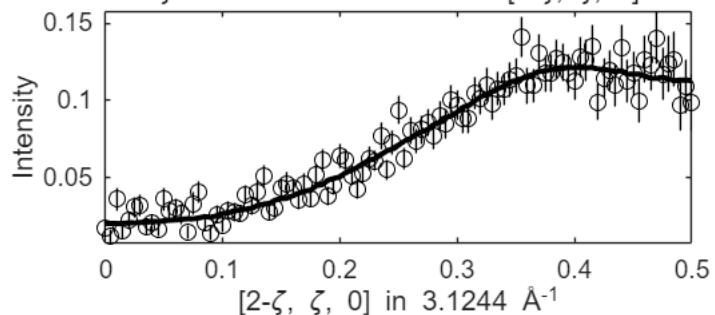
≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 248
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



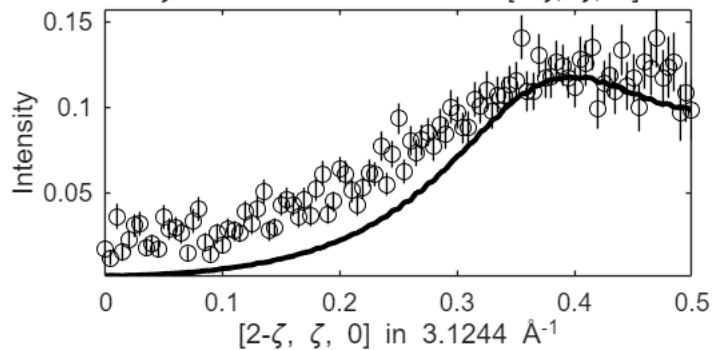
**** $E_n = 263 \pm 5$ going to 273

Fe_ei800.sqw

1 ± 0.13 ; $J_0 = 38 \pm 0.69$; $J_1 = 0 \pm 0$; $\gamma = 108.065 \pm$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 258
 $\zeta = -0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

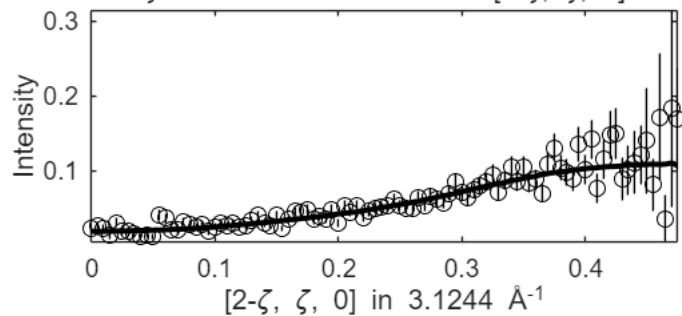


Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 258
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$

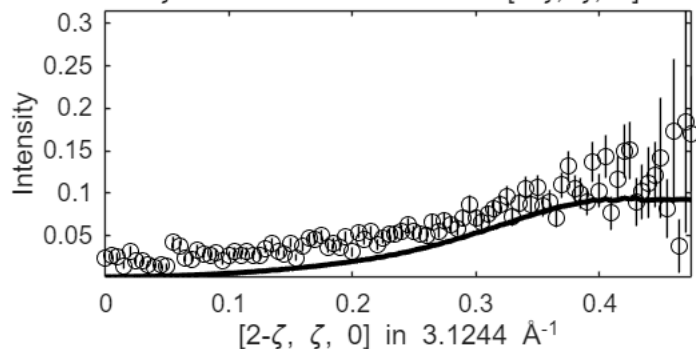


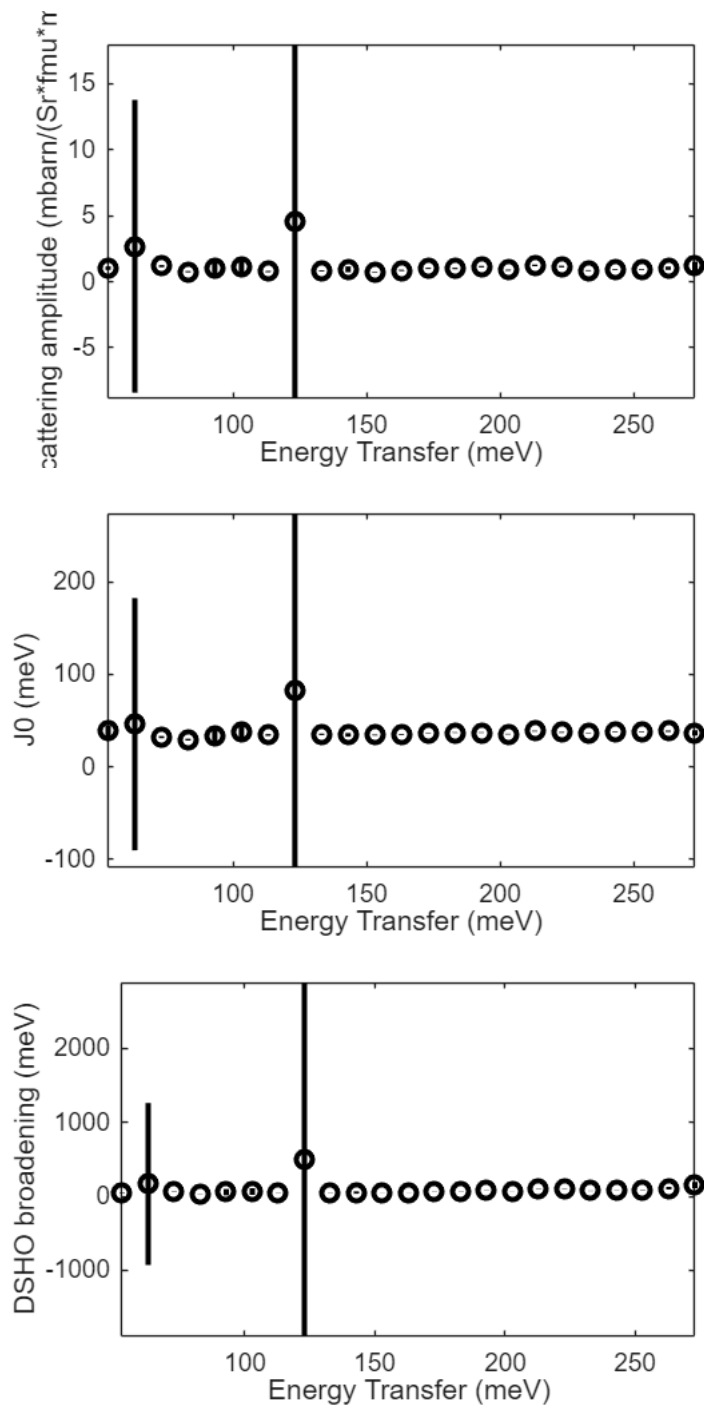
 **** En = 273±5 going to 273

Fe_ei800.sqw
 $.2 \pm 0.35$; $J_0 = 37 \pm 2.9$; $J_1 = 0 \pm 0$; $\gamma = 152.487 \pm 4$
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 268
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$



Fe_ei800.sqw
 ≤ 0.1 in $[2+\xi, \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 268
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, 0]$





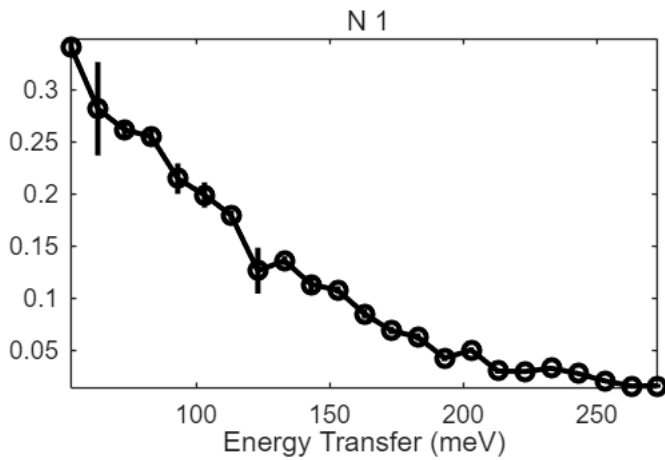
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

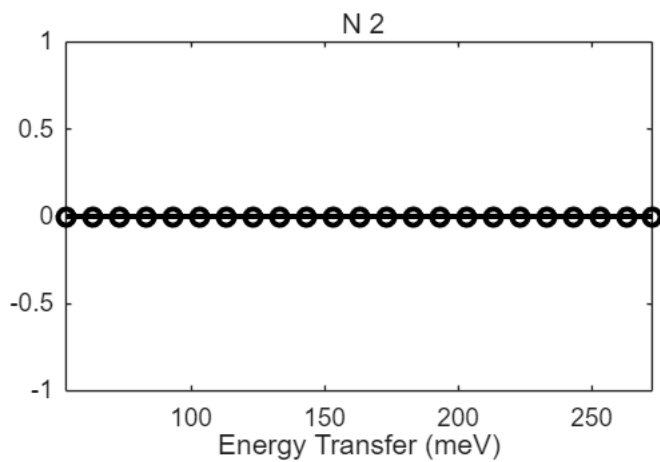
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```